DATA ANALYTICS • BUSINESS REPORT

WALMART RETAIL CUSTOMER RETENTION ANALYTICS

BUSINESS INSIGHTS REPORT

A data-driven review of customer churn, customer value, loyalty engagement, and retail revenue.

PREPARED BY Shubham Sharma	**TOOLS** SQL Server • Power BI
PROJECT DOMAIN Retail Customer Analytics	**REPORT SCOPE** 6 business questions

KEY FINDINGS

49.67%	60.00%	$134,282.18
Overall churn	Highest regional churn (West)	Top category revenue (Groceries)

Customer retention • Customer value • Loyalty • Revenue

1. Executive Summary

This project analyzes customer churn, customer lifetime value (CLV), loyalty engagement, and revenue patterns using SQL Server and Power BI. The analysis compares churn across regions and age groups, examines revenue by product category, reviews customer value segments, and compares loyalty points earned and redeemed. The findings highlight areas Walmart could investigate to improve retention, encourage repeat purchases, and strengthen loyalty-program engagement.

2. Overall Customer Churn Rate

Question: What proportion of customers are classified as churned?

SQL result: 149 of 300 customers were classified as churned, giving an overall churn rate of approximately **49.67%**.

Interpretation: Approximately half of the customers in the dataset are classified as churned.

Business implication: Walmart could investigate customer purchasing patterns and engagement to identify potential opportunities to improve customer retention.

3. Churn by Region

Question: Which region has the highest churn rate?

SQL result: The **West** region has the highest churn rate at **60.00%** (30 churned customers out of 50). Central follows at 58.33%, North at 49.18%, East at 43.10%, and South at 40.85%.

Interpretation: The West region has the highest observed churn rate, while South has the lowest among the regions shown.

Business implication: Investigate customer experience, engagement, and purchasing patterns in the West region to identify potential reasons for higher churn and develop targeted retention strategies.

Regional Churn Comparison

Region	Customers	Churned	Churn rate
West	50	30	60.00%
Central	60	35	58.33%
North	61	30	49.18%
East	58	25	43.10%
South	71	29	40.85%

4. Churn by Age Group

Question: Which age group has the highest churn rate?

SQL result: The **21–40** age group has the highest churn rate at **59.48%**, with 69 of 116 customers classified as churned. The 0–20 group is at 46.67%, the 40–60 group at 43.55%, and the 61+ group at 42.22%.

Interpretation: The 21–40 age group has the highest churn rate in the supplied results. The 61+ group has the lowest.

Business implication: Walmart could investigate the purchasing patterns and engagement of customers aged 21–40 and test targeted retention strategies to encourage repeat purchases.

Age-Group Churn Comparison

Age group	Customers	Churned	Churn rate
21–40	116	69	59.48%
0–20	15	7	46.67%
40–60	124	54	43.55%
61+	45	19	42.22%

5. Revenue Analysis

Question: Which product category contributes the most revenue?

SQL result: Groceries generated the highest revenue at **$134,282.18**, contributing approximately **26.01%** of total revenue.

Interpretation: Groceries is the top revenue-generating category, followed by Electronics (25.28%), Home & Living (24.91%), and Apparel (23.80%).

Business implication: Walmart could investigate opportunities to encourage repeat purchases in Groceries and explore cross-selling opportunities with other categories.

Revenue by Product Category

Product category	Total revenue (USD)	Contribution
Groceries	$134,282.18	26.01%
Electronics	$130,494.64	25.28%
Home & Living	$128,621.76	24.91%
Apparel	$122,861.94	23.80%

Revenue insight

Groceries leads the four listed categories, but the revenue contributions are relatively close (about 23.80%–26.01%). This suggests revenue is spread across categories rather than dominated by a single category.

6. Customer Lifetime Value (CLV) Analysis

Question: How are customers distributed across CLV segments?

Power BI result: The dashboard shows **208 Low-CLV customers** and **92 High-CLV customers**, for 300 customers in total.

Interpretation: Low-CLV customers represent approximately 69.33% of the total, while High-CLV customers account for approximately 30.67%. The chart shows customer counts; average CLV values were not displayed.

Business implication: Walmart could prioritize retention initiatives for high-CLV customers who show signs of inactivity while exploring ways to increase repeat purchases and customer value among low-CLV customers.

CLV Segment Distribution

Segment	Customers	Share of total
Low CLV	208	69.33%
High CLV	92	30.67%

7. Loyalty Program

Question: How do earned and redeemed loyalty points compare?

Power BI result: Premium earned approximately 0.20M and redeemed 0.17M points; Plus earned 0.20M and redeemed 0.18M; Basic earned 0.18M and redeemed 0.12M; Elite earned 0.18M and redeemed 0.15M.

Interpretation: Plus has the highest displayed redemption level (about 0.18M points), while Basic has the lowest (about 0.12M). Earned points exceed redeemed points across all four tiers.

Business implication: Walmart could investigate whether clearer loyalty-program communication and more convenient redemption options would encourage members, particularly Basic members, to redeem more of their earned points.

Points Earned vs. Redeemed by Loyalty Tier

Loyalty tier	Points earned	Points redeemed
Premium	0.20M	0.17M
Plus	0.20M	0.18M
Basic	0.18M	0.12M
Elite	0.18M	0.15M

8. Limitations

The analysis depends on the quality and completeness of the supplied dataset. Observed relationships do not establish causation. Loyalty-point values are rounded in the Power BI chart, so exact totals and redemption percentages should be calculated from the underlying data before reporting them precisely. Business recommendations should be validated through further analysis or testing.

Prepared by: Shubham Sharma
Tools: SQL Server and Power BI

Report focus
Retention • CLV • Loyalty • Revenue